TCC805x MCU BSP

API Specification

for General Purpose Serial Bus Controller

Rev. 1.01 [A]

2021-01-22

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1 INTRODUCTION

This document describes the general purpose serial bus controller device driver (hereinafter referred to as GPSB driver) of TCC805x MCU BSP. For more detailed information of the general purpose serial bus controller in the TCC805x MCU Sub-system, refer to Chapter 11 General Purpose Serial Bus (GPSB) Controller of the ***"PART 11-MICOM SUB-SYSTEM"*** in the ***"TCC805x Full Specification"***. [1]

2 TERMS AND ABBREVIATIONS

Table 2.1 Terms and Abbreviation

GPSB	General Purpose Serial Bus
BPW	Bit Per Width
SCK	GPSB(SPI) Clock
SDO	GPSB(SPI) Output (Master channel output, Slave channel is not supported.)
SDI	GPSB(SPI) Input (Master channel input, Slave channel is not supported.)
CS	Chip Select PIN of GPSB
SDM	Safe Display Manager
RVC	Rear Video Camera

3 SOURCE FILES

3.1 Location of Source Files

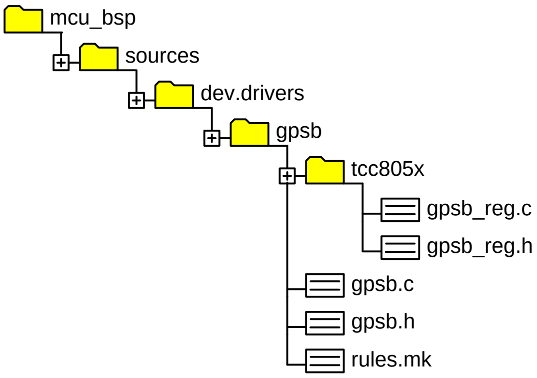

Figure 3.1 Location of the source files

3.2 Simple Description of Source Files

- gpsb_reg.c
  - GPSB specific source code by chipset.
- gpsb_reg.h
  - GPSB specific header by chipset.
- gpsb.c
  - GPSB common source code.
- gpsb.h
  - GPSB common header including API & DEFINE for user
- rules.mk
  - Rules used for build

4 CONSTANTS

4.1 GPSB Base Address

Constants values for base register address for each GPSB channel and GPSB configuration. They are used to change register value for the GPSB operation.

Declaration

```
#define TCC_MICOM_GPSB_BASE
#define GPSB0_BASE          (TCC_MICOM_GPSB_BASE + 0x00000)
#define GPSB1_BASE          (TCC_MICOM_GPSB_BASE + 0x10000)
#define GPSB2_BASE          (TCC_MICOM_GPSB_BASE + 0x20000)
#define GPSB3_BASE          (TCC_MICOM_GPSB_BASE + 0x30000)
#define GPSB4_BASE          (TCC_MICOM_GPSB_BASE + 0x40000)
#define GPSB5_BASE          (TCC_MICOM_GPSB_BASE + 0x50000)
#define GPSB_PCFG_BASE      (TCC_MICOM_GPSB_BASE + 0x60000)
#define GPSB_IO_MONITOR_BASE (TCC_MICOM_GPSB_BASE + 0x70000)
```

Description

Value	Description
TCC_MICOM_GPSB_BASE	GPSB controller register base address: 0x1B100000UL
GPSB0_BASE	GPSB controller channel 0 base address: 0x1B100000UL
GPSB1_BASE	GPSB controller channel 1 base address: 0x1B110000UL
GPSB2_BASE	GPSB controller channel 2 base address: 0x1B120000UL
GPSB3_BASE	GPSB controller channel 3 base address: 0x1B130000UL
GPSB4_BASE	GPSB controller channel 4 base address: 0x1B140000UL
GPSB5_BASE	GPSB controller channel 5 base address: 0x1B150000UL
GPSB_PCFG_BASE	GPSB controller port configuration base address: 0x1B160000UL
GPSB_IO_MONITOR_BASE	GPSB controller IO monitor base address: 0x1B170000UL

Remark

None

4.2 GPSB Channel

Constants values for index of each channel.

Declaration

```
#define GPSB_CH_0     (0UL)
#define GPSB_CH_1     (1UL)
#define GPSB_CH_2     (2UL)
#define GPSB_CH_3     (3UL)
#define GPSB_CH_4     (4UL)
```

Description

Value	*Description*
GPSB_CH_0	*GPSB Channel 0 index value*
GPSB_CH_1	*GPSB Channel 1 index value*
GPSB_CH_2	*GPSB Channel 2 index value*
GPSB_CH_3	*GPSB Channel 3 index value. (This channel is reserved for SDM. If you want to use this channel for normal purpose, change SDM channel value to 0x6 or 0x7 via calling API GPSB_SetSdmChaanel.)*
GPSB_CH_4	*GPSB Channel 4 index value. (This channel is reserved for SRVC. If you want to use this channel for normal purpose, change SDM channel value to 0x6 or 0x7 via calling API GPSB_SetRvcChaanel.)*

Remark

Refer to ***SDM_RVC_CTRL_SEL*** in "*TCC805x Full Specification*". [1]

4.3 GPSB Mode

Constants values for the GPSB transfer mode as parameter of *GPSB_Xfer* function.

Declaration

```
#define GPSB_XFER_MODE_WITHOUT_INTERRUPT    (uint32)(0x01UL)
#define GPSB_XFER_MODE_WITH_INTERRUPT       (uint32)(0x02UL)
#define GPSB_XFER_MODE_DMA                  (uint32)(0x10UL)
#define GPSB_XFER_MODE_PIO                  (uint32)(0x20UL)
#define GPSB_XFER_MODE_WITHOUT_CTF          (uint32)(0x40UL)
```

Description

Value	Description
GPSB_XFER_MODE_WITHOUT_INTERRUPT	Non-interrupt transfer mode
GPSB_XFER_MODE_WITH_INTERRUPT	Interrupt transfer mode
GPSB_XFER_MODE_DMA	DMA transfer mode
GPSB_XFER_MODE_PIO	PIO (FIFO) transfer mode
GPSB_XFER_MODE_WITHOUT_CTF	Continuous transfer mode (Keep CS signal activate state)

Remark

None

4.4 GPSB Event

Constants values for the GPSB error event. When the GPSB error interrupt occurs, the GPSB driver notifies a user of the errors via callback function.

Declaration

```
#define GPSB_EVENT_COMPLETE   (uint32)(0x0000UL)
#define GPSB_EVENT_ERR_ROR    (uint32)(0x0100UL)
#define GPSB_EVENT_ERR_WUR    (uint32)(0x0200UL)
#define GPSB_EVENT_ERR_RUR    (uint32)(0x0400UL)
#define GPSB_EVENT_ERR_WOR    (uint32)(0x0800UL)
```

Description

Value	Description
GPSB_EVENT_COMPLETE	Transfer complete
GPSB_EVENT_ERR_ROR	Read FIFO over-run error
GPSB_EVENT_ERR_WUR	Write FIFO under-run error
GPSB_EVENT_ERR_RUR	Read FIFO under-run error
GPSB_EVENT_ERR_WOR	Write FIFO over-run error

Remark

None

4.5 GPSB State

Constants values for the GPSB transfer state management.

Declaration

```
#define GPSB_XFER_STATE_DISABLED     (uint8)(0x00UL)
#define GPSB_XFER_STATE_IDLE         (uint8)(0x01UL)
#define GPSB_XFER_STATE_RUNNING      (uint8)(0x02UL)
```

Description

Value	Description
GPSB_XFER_STATE_DISABLED	GPSB is disabled.
GPSB_XFER_STATE_IDLE	GPSB is in idle state.
GPSB_XFER_STATE_RUNNING	GPSB is in running state. Other transfer requests are blocked in this state.

Remark

None

5 STRUCTURES

5.1 GPSBDev_t

Structure to store operating information of each GPSB channel. GPSBDev_t is declared in ***gpsb_reg.c*** and has a default initial value. If you input parameter values when calling API such as ***GPSB_Open***, the parameter values are automatically stored in this struct member variable.

Declaration

```
typedef struct _GPSBDev_
{
    Uint32              dChannel;
    uint32              dBase;
    uint32              dCfgAddr;
    uint32              dPeriName;
    uint32              dIobusName;
    uint32              dIsSlave;
    uint32              dSdo;
    uint32              dSdi;
    uint32              dSclk;
    uint32              dPort;
    uint32              dIntrPriority;
    uint32              dSupportDma;
    GPSBDmaBuff_t       dTxDma;
    GPSBDmaBuff_t       dRxDma;
    const void *        dAsyncTxBuf;
    void *              dAsyncRxBuf;
    uint32              dAsyncTxDataSize;
    uint32              dAsyncRxDataSize;
    uint8               dState;
    GPSBXferComplite_t  dComplete;
} GPSBDev_t;
```

Description

Value	Description
dChannel	GPSB channel number
dBase	GPSB channel base address
dCfgAddr	GPSB port configuration address
dPeriName	Peri clock index
dIobusName	IOBUS index
dIsSlave	GPSB mode (Master or Slave)
dSdo	SDO(MOSI) GPIO index info.
dSdi	SDI(MISO) GPIO index info.
dSclk	SCLK GPIO index info.
dPort	Matched port number
dIntrPriority	Interrupt priority
dSupportDma	Support info. Default: FALSE
dTxDma	Tx DMA Buffer info.
dRxDma	Rx DMA Buffer info.
dAsyncTxBuf	Tx buffer address for FIFO async transfer
dAsyncTxBuf	Rx buffer address for FIFO async transfer
dAsyncTxDataSize	Async FIFO transfer Tx data size
dAsyncRxDataSize	Async FIFO transfer Rx data size
dState	State of GPSB
dComplete	Notify the completion of non-blocking transfer.

Remark

None

5.2 GPSBXferComplete_t

Structure to register callback function. The value of ***xcCallback*** parameters is filled when calling ***GPSB_Open***.

Declaration

```
typedef struct _GPSBXferComplete_
{
    GPSBCallback        xcCallback;
    void *              xcArg;
} GPSBXferComplite_t;
```

Description

Value	*Description*
xcCallback	*Callback function pointer. Refer to GPSBCallback.*
xcArg	*Etc.*

Remark

None

6 CALLBACK FUNCTIONS

6.1 GPSBCallback

When async transfer is completed or an error interrupt occurs, the GPSB driver notifies a user of the event information via callback function.

The supported events are as follows:

- Transmit done.
- Error interrupts.

Declaration

```
typedef void (*GPSBCallback)
(
    uint32      uiCh,
    int32       iEvent,
    void *      pArg
);
```

Parameters

Value	Description
uiCh	Value of channel to be controlled
iEvent	Value of Event information. Refer to GPSB Event.
pArg	Parameter from user

Return

None

Remark

None

7 APIs

7.1 GPSB_Init

Registers the GPSB interrupt handler functions and enables the GPSB interrupts. Normally, interrupts occur when using FIFO or DMA async transfer.
Supported interrupts: GPSB 0~4, GPSB DMA 0~4

Declaration

```
void GPSB_Init
(
    void
);
```

Parameters

None

Return

None

Remark

None

7.2 GPSB_Deinit

Disables all the GPSB interrupts.

Declaration

```
void GPSB_Deinit
(
    void
);
```

Parameters

None

Return

None

Remark

None

7.3 GPSB_Reset

Resets the GPSB channel in IOBUS. After reset, the GPSB channel register is set to default value.

Declaration

```
SALRetCode_t GPSB_Reset
(
    Uint32     uiCh
);
```

Parameters

Value	Description
uiCh	Value of channel to be controlled

Return

Value	Description
SAL_RET_SUCCESS	Success
SAL_RET_FAILED	Failure

Remark

None

7.4 GPSB_Open

Initializes the GPSB channel. Input valid parameter to operate the GPSB channel. SCK/SDO/SDI values depend on HW construction. (If use internal path such as loopback mode, can set SDO/SDI/SCK to 0.) If want to get notification from the GPSB driver, register callback function.

Declaration

```
SALRetCode_t GPSB_Open
(
    uint32          uiCh,
    uint32          uiSdo,
    uint32          uiSdi,
    uint32          uiSclk,
    uint32          uiDmaAddrTx,
    uint32          uiDmaAddrRx,
    uint32          uiDmaBufSize,
    GPSBCallback    fbCallback,
    void *          pArg
);
```

Parameters

Value	Description
uiCh	Value of channel to be controlled
uiSdo	SDO GPIO index
uiSdi	SDI GPIO index
uiSclk	SCLK GPIO index
uiDmaAddrTx	DMA TX buffer physical address
uiDmaAddrRx	DMA RX buffer physical address
uiDmaBufSize	DMA buffer size
fbCallback	Callback function pointer. Refer to the GPSBCallback.
pArg	Etc.

Return

Value	Description
SAL_RET_SUCCESS	Success
SAL_RET_FAILED	Failure

Remark

None

7.5 GPSB_Close

De-initializes GPSB channel.

- Stop DMA.
- Disable channel interrupt.
- Clear register value to disable GPSB channel.

Declaration

```
SALRetCode_t GPSB_Close
(
    uint32      uiCh
);
```

Parameters

Value	Description
uiCh	Value of channel to be controlled

Return

Value	Description
SAL_RET_SUCCESS	Success
SAL_RET_FAILED	Failure

Remark

None

7.6 GPSB_SetSpeed

Sets GPSB operating clock speed. The maximum operating clock speed is limited by HW construction and signal strength.

Declaration

```
SALRetCode_t GPSB_SetSpeed
(
    uint32      uiCh,
    uint32      uiSpeedInHz
);
```

Parameters

Value	Description
uiCh	Value of channel to be controlled
uiSpeedInHz	SCLK speed

Return

Value	Description
SAL_RET_SUCCESS	Success
SAL_RET_FAILED	Failure

Remark

None

7.7 GPSB_SetBpw

Sets the BPW value. Valid BPW value is 8, 16, or 32.

Declaration

```
SALRetCode_t GPSB_SetBpw
(
    uint32      uiCh,
    uint32      uiBpw
);
```

Parameters

Value	Description
uiCh	Value of channel to be controlled
uiBpw	Bit per word

Return

Value	Description
SAL_RET_SUCCESS	Success
SAL_RET_FAILED	Failure

Remark

None

7.8 GPSB_SetMode

Sets the GPSB operating mode. Polarity and phase of the GPSB signal change according to the mode value. Refer to GPSB MODE register & Timing Diagram in *"TCC805x Full Specification"*.

Declaration

```
void GPSB_SetMode
(
    uint32      uiCh,
    uint32      uiMode
);
```

Parameters

Value	Description
uiCh	Value of channel to be controlled
uiMode	Operating Mode. (e.g. GPSB_CPOL, GPSB_CPHA, GPSB_LOOP, etc.)

Return

None

Remark

None

7.9 GPSB_CsInit

Initializes CS GPIO to default config.

Declaration

```
SALRetCode_t GPSB_CsInit
(
    uint32      uiCh,
    uint32      uiCs,
    uint32      uiCsHigh
);
```

Parameters

Value	Description
uiCh	Value of channel to be controlled
uiCs	CS GPIO index
uiCsHigh	Default high or low

Return

Value	Description
SAL_RET_SUCCESS	Success
SAL_RET_FAILED	Failure

Remark

None

7.10 GPSB_CsAlloc

Configures the CS GPIO.

Declaration

```
SALRetCode_t GPSB_CsAlloc
(
    uint32     uiCh,
    uint32     uiCs
);
```

Parameters

Value	Description
uiCh	Value of channel to be controlled
uiCs	CS GPIO index

Return

Value	Description
SAL_RET_SUCCESS	Success
SAL_RET_FAILED	Failure

Remark

None

7.11 GPSB_CsActivate

Activates the CS GPIO.

Declaration

```
SALRetCode_t GPSB_CsActivate
(
    uint32     uiCh,
    uint32     uiCs,
    uint32     uiCsHigh
);
```

Parameters

Value	Description
uiCh	Value of channel to be controlled
uiCs	CS GPIO index
uiCsHigh	Default High or Low

Return

Value	Description
SAL_RET_SUCCESS	Success
SAL_RET_FAILED	Failure

Remark

None

7.12 GPSB_CsDeactivate

Deactivates the CS GPIO.

Declaration

```
SALRetCode_t GPSB_CsDeactivate
(
    uint32      uiCh,
    uint32      uiCs,
    uint32      uiCsHigh
);
```

Parameters

Value	Description
uiCh	Value of channel to be controlled
uiCs	CS GPIO index
uiCsHigh	Default High or Low

Return

Value	Description
SAL_RET_SUCCESS	Success
SAL_RET_FAILED	Failure

Remark

None

7.13 GPSB_Xfer

Transmits data. This function can be called from application. Before calling this function, the GPSB channel must set initializing and configuration (speed, BPW, etc.).

Declaration

```
SALRetCode_t GPSB_Xfer
(
    uint32          uiCh,
    const void *    pTx,
    void *          pRx,
    uint32          uiDataSize,
    uint32          uiXferMode
);
```

Parameters

Value	Description
uiCh	Value of channel to be controlled
pTx	TX data buffer
pRx	RX data buffer
uiDataSize	Write/Read data size
uiXferMode	Transfer mode (DMA/PIO, Interrupt/Non-Interrupt)

Return

Value	Description
SAL_RET_SUCCESS	Success
SAL_RET_FAILED	Failure

Remark

None

7.14 GPSB_XferAbort

Stops transmission.

Declaration

```
void GPSB_XferAbort
(
    uint32      uiCh
);
```

Parameters

Value	Description
uiCh	Value of channel to be controlled

Return

Value	Description
SAL_RET_SUCCESS	Success
SAL_RET_FAILED	Failure

Remark

None

7.15 GPSB_SetSdmChannel

Sets SDM channel. The SDM default channel is ***GPSB_CH_3***.

Declaration

```
SALRetCode_t GPSB_SetSdmChannel
(
    uint32      uiCh
);
```

Parameters

Value	***Description***
uiCh	*Value of channel to be controlled*

Return

Value	***Description***
SAL_RET_SUCCESS	*Success*
SAL_RET_FAILED	*Failure*

Remark

None

7.16 GPSB_SetRvcChannel

Sets RVC channel. The default channel of RVC is ***GPSB_CH_4***.

Declaration

```
SALRetCode_t GPSB_SetRvcChannel
(
    uint32      uiCh
);
```

Parameters

Value	*Description*
uiCh	*Value of channel to be controlled*

Return

Value	*Description*
SAL_RET_SUCCESS	*Success*
SAL_RET_FAILED	*Failure*

Remark

None

8 How to use GPSB Driver

GPSB driver requires channel initializing & de-initializing step before and after the data transfer.

[Initialize]

Normally, you can set up channels via calling ***GPSB_Open*** with valid parameters. If external path is used, <u>not</u> internal path such as loopback, you should set SCK, SDI, SDO, and CS GPIO. Operating values such as speed, Bit Per Word (BPW), and mode can be set after completing ***GPSB_Open*** without error.

[Data transfer]

After initializing, data can be transferred via calling ***GPSB_Xfer***. ***GPSB_Xfer*** requires valid buffer and xfer mode information. Before and after calling ***GPSB_Xfer***, if CTF mode is not used, you should set CS GPIO to control xfer start and stop. Then call ***GPSB_CsActivate*** & ***GPSB_CsDeactivate*** to control the CS GPIO.

[De-initialize]

After the use of GPSB channel is completed, close GPSB channel via calling ***GPSB_Close*** to prevent useless resource and to avoid errors caused by duplicated resource.

8.1 Operation of GPSB

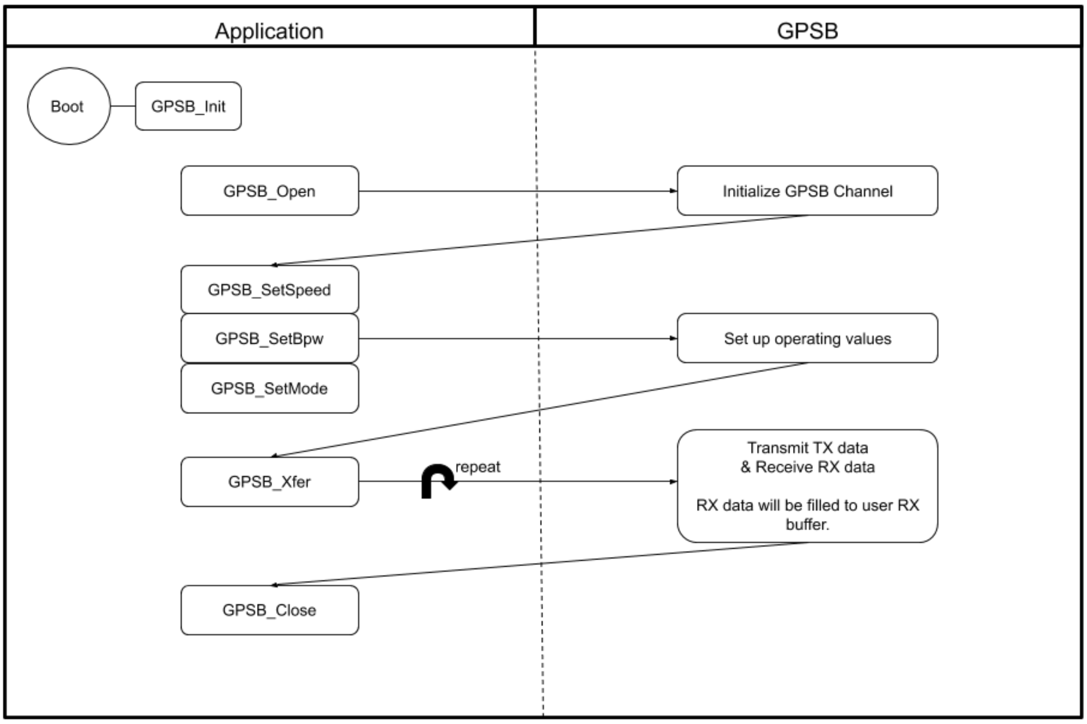

Figure 8.1 Diagram for GPSB

8.2 Example Code

Refer to test code in ***gpsb_test.c*** file.

```
static uint32     g_tx_dma;
static uint32     g_rx_dma;
static uint8      g_tx[GPSB_TEST_BUF_SIZE]  = {0,};
static uint8      g_rx[GPSB_TEST_BUF_SIZE]  = {0,};

{
    ucCh            = 0UL;
    dma_buf_size    = DMA_BUF_SIZE;
    gpio_sclk       = GPIO_GPB(25UL);
    gpio_cs         = GPIO_GPB(26UL);
    gpio_sdo        = GPIO_GPB(27UL);
    gpio_sdi        = GPIO_GPB(28UL);
    speed           = 20*1000*1000;
    ucBpw           = 16;
    cs_high         = 0;
    xfer_mode       = GPSB_XFER_MODE_WITHOUT_INTERRUPT;

    // initialize channel
    GPSB_Open(ucCh, gpio_sdo, gpio_sdi, gpio_sclk, g_tx_dma, g_rx_dma,
              dma_buf_size, NULL, NULL_PTR);

    // configuration
    GPSB_SetSpeed(ucCh, speed);
    GPSB_SetBpw(ucCh, ucBpw);
    GPSB_SetMode(ucCh, 0);

    // Set TX data in buffer (g_tx)
    GPSB_CsActivate(ucCh, gpio_cs, cs_high);
    GPSB_Xfer(ucCh, g_tx, g_rx, data_size, xfer_mode);
    GPSB_CsDeactivate(ucCh, gpio_cs, cs_high);

    // Check RX data in buffer (g_rx)

    // Disable channel
    GPSB_Close(ucCh);
}
```

9 REFERENCES

[1] TCC805x Full Specification

[2] Contact Telechips for more details: sales@telechips.com

10 Revision History

Rev. 1.01: 2021-01-22

- Updated
    - Chapter 7.4: Declaration

Rev. 1.00: 2020-11-12

- First version release